Generative AI & Prompt Engineering Internship

Skillify



Task 2

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Submitted to

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Task 2: Student Performance Analyzer:

Objective

Create a Python program that analyzes student information, calculates grades and averages, identifies performance levels, and generates a summary.

Requirements

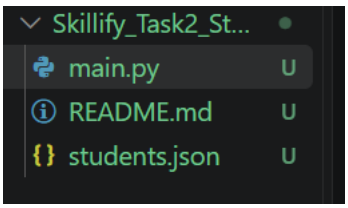
- Python
- Lists
- Tuples
- Dictionaries
- Functions
- User input
- File handling
- Exception handling
- Basic data processing
- Modules

Submission Format

- submit a pdf containing followings:
- Python source code
- Sample student data
- Output screenshots

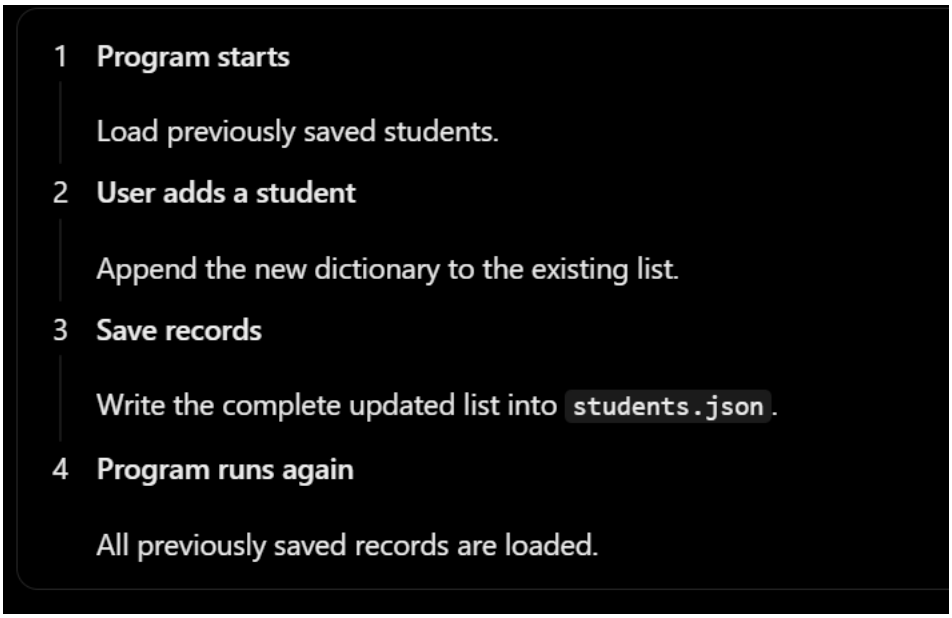
Solution:

First I built a basic structure of my project:



Main.py file is for writing the code of performance analyzer, README.md file for describing the technologies & task, how to run the file etc and as for students.json it stores the student data.

The logic is built here:



First I built a tuple that contains the subjects then I coded a empty list of student that will store the students dictionary and then I coded a add_student function where I take the name,Rollno,subject marks as input and append(save) them into the student list I coded earlier then I coded a calculate_average(marks) function which takes the students marks sums them and returns the avg of it. Then I coded another function named calculate_grade(average) which takes avg and compare

it and grade it according A+,A,B...F , there is one more function that calculate performance level into excellent ,good, bad…etc and this function name is get_performance_level(average) and here main functionalities of my student completed now I makes my terminal output more presentable also added a student summary ,then I move on file handling for it I imported “json” library tat stores the data in json format and use my students.json file to store it

I added new function save_students_to_file() which uses ‘w’ write mode to write in file opens it and writes in it the user name,Rollno,marks of each subject.

Then I added new function to load the data from file and then appends the new data into it the function named is load_students_from_file().

```
STUDENT PERFORMANCE ANALYZER
----------------------------
Available Subjects: ('Mathematics', 'English', 'Computer Science')
Enter student name: Ali
Enter student roll number: 245
Mathematics
Enter marks for Mathematics: 30
English
Enter marks for English: 60
Computer Science
Enter marks for Computer Science: 57
Student added successfully!
========== STUDENT SUMMARY ==========
Name: Ali
Roll Number: 245

Subject-wise Marks:
Mathematics: 30
English: 60
Computer Science: 57

Total Marks: 147
Average: 49.00
Grade: F
Performance: Poor

=======================================
```

```
STUDENT PERFORMANCE ANALYZER
----------------------------
Available Subjects: ('Mathematics', 'English', 'Computer Science')
Enter student name: Ahmad
Enter student roll number: 256
Mathematics
Enter marks for Mathematics: 50
English
Enter marks for English: 70
Computer Science
Enter marks for Computer Science: 80
Student added successfully!
========== STUDENT SUMMARY ==========
Name: Ahmad
Roll Number: 256

Subject-wise Marks:
Mathematics: 50
English: 70
Computer Science: 80

Total Marks: 200
Average: 66.67
Grade: C
Performance: Average

=======================================
```

```
[
    {
        "name": "Ali",
        "roll_no": 245,
        "marks": {
            "Mathematics": 30,
            "English": 60,
            "Computer Science": 57
        }
    },
    {
        "name": "Ahmad",
        "roll_no": 256,
        "marks": {
            "Mathematics": 50,
            "English": 70,
            "Computer Science": 80
        }
    }
]
```

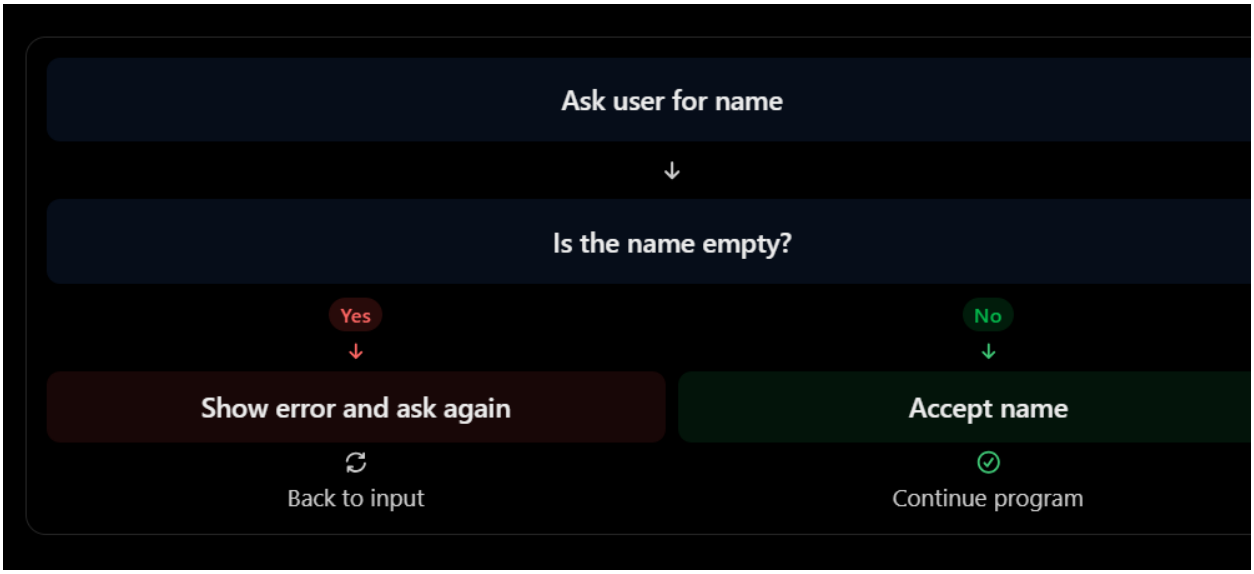

Then I added exception handling start it with validating the Rollno that it should be unique and only integer no chars are allowed show error and alke for input Rollno again.

```
Available Subjects: ('Mathematics', 'English', 'Computer Science')
Enter student name: mahi
Enter student roll number: 78u
Error: Invalid roll number. Please enter a valid integer.
Enter student roll number: 123
Mathematics
```

Then I added exception handling at marks to check marks is within the range of 0 and 100 and only integers no characters are allowed.

```
Enter student roll number: 457
Mathematics
Enter marks for Mathematics: 12s
Error: Invalid marks. Please enter a valid integer.
Enter marks for Mathematics: 110
Error: Marks must be between 0 and 100.
Enter marks for Mathematics: -4
Error: Marks must be between 0 and 100.
Enter marks for Mathematics: 24
English
```

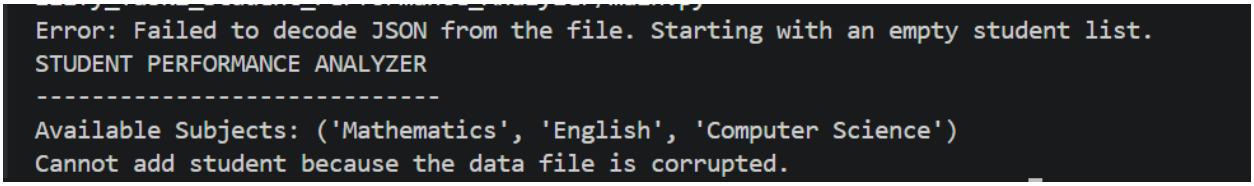
Then I added exception handling at name to check if it is empty and also added .strip() to remove any white spaces in start or end of the name.



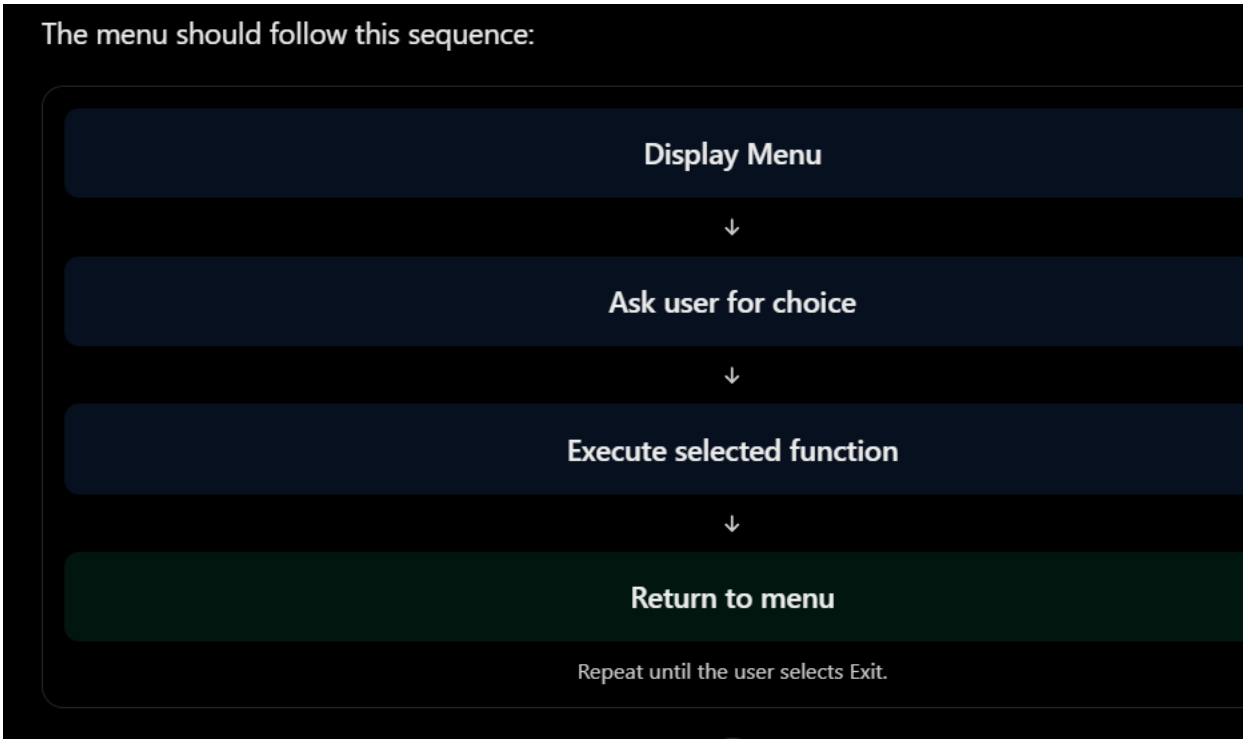

```
STUDENT PERFORMANCE ANALYZER
----------------------------
Available Subjects: ('Mathematics', 'English', 'Computer Science')
Enter student name:
Error: Name cannot be empty. Please enter a valid name.
Enter student name:
Error: Name cannot be empty. Please enter a valid name.
Enter student name: nm
Enter student roll number: Traceback (most recent call last):
```

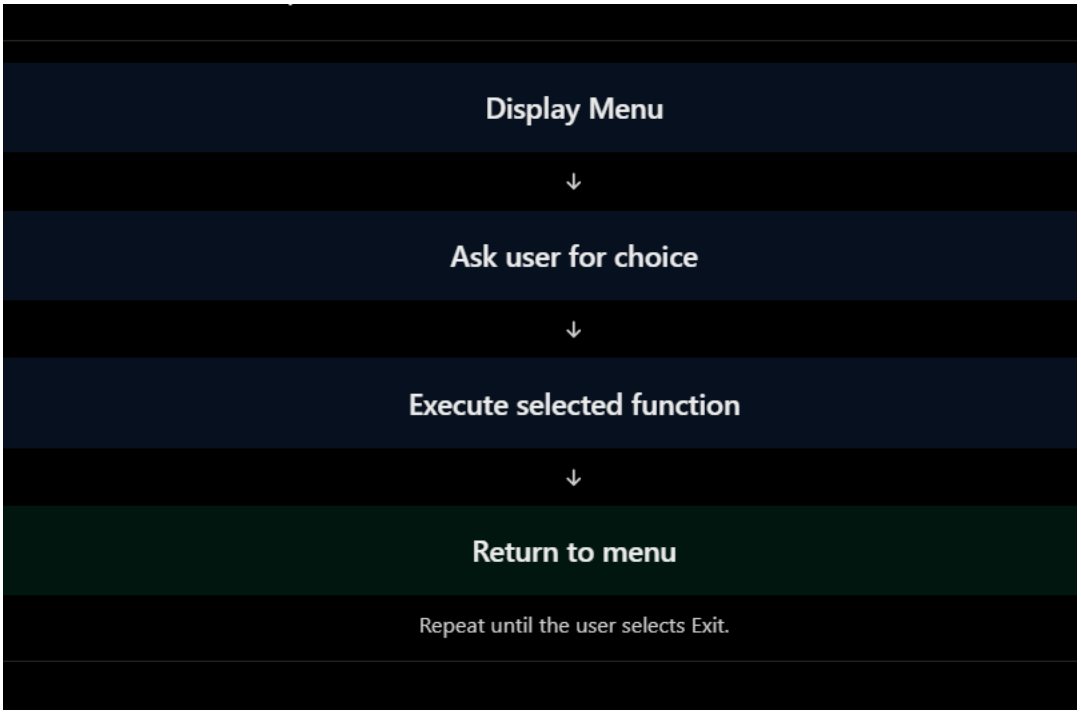
Then I added Exception Handling upon loading data and saving data in file

I tested by making a small change in .json file by removing a comma



Now its time to add Menu:





After adding menu and polishing it :

```
======STUDENT PERFORMANCE ANALYZER=======
Menu:
1. Add Student
2. Display all Students Record
3. Exit
Enter the option: e
Error: Invalid option. Please enter a valid integer option.
Menu:
1. Add Student
2. Display all Students Record
3. Exit
Enter the option: 3
```

```
 ======STUDENT PERFORMANCE ANALYZER=======
 Menu:
 1. Add Student
 2. Display all Students Record
 3. Exit
 Enter the option: 2

 ========== ALL STUDENT RECORDS =========
 ----------------------------------------
 Name: mn
 Roll Number: 89
 Subject-wise Marks:
 Mathematics: 67
 English: 45
 Computer Science: 23
 Average: 45.00
 Grade: F
 ----------------------------------------
 Name: Nayab
 Roll Number: 3015
 Subject-wise Marks:
 Mathematics: 67
 English: 78
```

```
========== ADD STUDENT ===============
Enter student name: Amar
Enter student roll number: 235
Mathematics
Enter marks for Mathematics: 12
English
Enter marks for English: 89
Computer Science
Enter marks for Computer Science: 67
Student added successfully!
==================END====================
========== STUDENT SUMMARY ==========
Name: Amar
Roll Number: 235
Subject-wise Marks:
Mathematics: 12
English: 89
Computer Science: 67
Total: 168
Average: 56.00
Grade: D
Performance: Below Average
```